Lab 5
Thevenin Equivalents of Lab Equipment

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I. Introduction

This lab had two main objectives. The first was to find the oscilloscope's input resistance using two different methods: a direct ohmmeter reading, and a voltage-division measurement using a DC power supply and a series resistor. The second was to find the Thevenin equivalent of the function generator at 1 kHz by measuring its open-circuit voltage and its voltage under a known load. These two equivalent-circuit models were then used to calculate how much loading occurs when the oscilloscope is connected to the function generator, and to find the general condition on input resistance needed to keep that loading error under 1%.

II. Results

1. Oscilloscope Input Resistance

The oscilloscope's DC input resistance was first measured directly with a DMM across the BNC center pin and ground, with the oscilloscope set to DC coupling. This was measured to be 0.2059 MΩ

The input resistance was then found a second way, using the setup in Fig. 1: a DC power supply (Vs) in series with a resistor R1 comparable to the oscilloscope's input resistance (300 kΩ-3 MΩ), with the oscilloscope used to measure the voltage across itself.

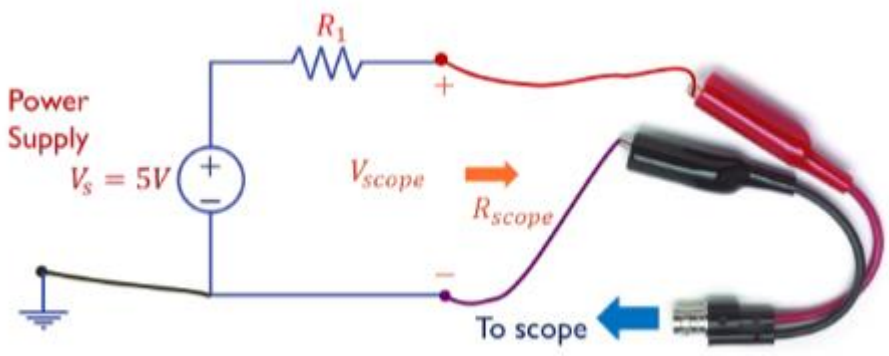


Figure 1: Setup for measuring Thevenin's equivalent resistance of the oscilloscope [Ref. 1]

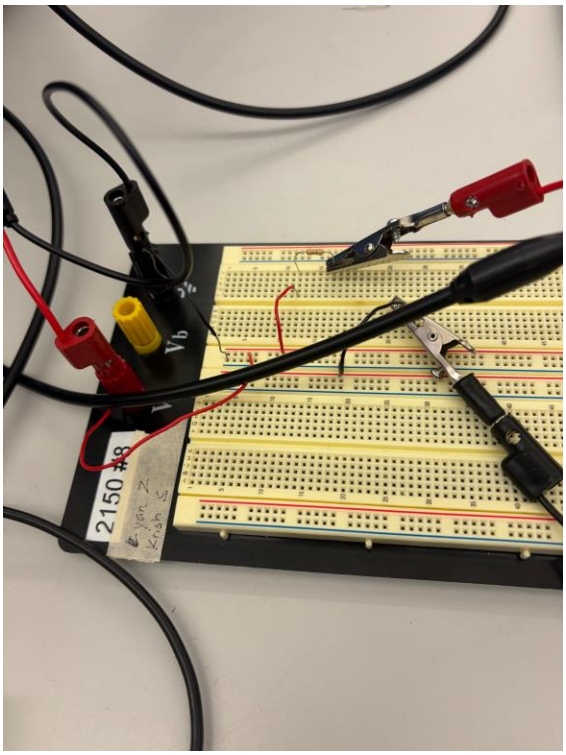


Figure 2: Fig.1 built on a breadboard

Using the measured resistance R1(0.975MΩ), the power supply voltage Vs(5.00V), and the voltage Vscope measured at the oscilloscope(2.45V), the input resistance follows from the voltage-divider relationship: Vscope = Vs × (Rin / (R1 + Rin)), which rearranges to Rin = R1 × Vscope / (Vs − Vscope). The DC input resistance was found to be 0.937MΩ

The ohmmeter reading (0.2059 MΩ) differed substantially from the voltage-division result (≈0.938 MΩ), a difference far larger than expected for two methods measuring the same resistance. This highlights a procedural issue rather than normal measurement tolerance. The oscilloscope was likely still connected to the rest of the circuit when the ohmmeter reading was taken, meaning the meter measured Rin in parallel with the path containing R1 rather than Rin alone. This would read lower than the true value, consistent with what was observed. The voltage-division result is closer to the ~1 MΩ typical of a scope input and is treated as the more reliable of the two values for future discussion.

2. Thevenin Equivalent of the Function Generator

The function generator was set to 1 kHz and 1 Vpp. Its Thevenin equivalent (Fig. 3) was found by measuring the open-circuit voltage Voc, then the voltage across a 100 Ω load resistor (VrL) connected to the output.

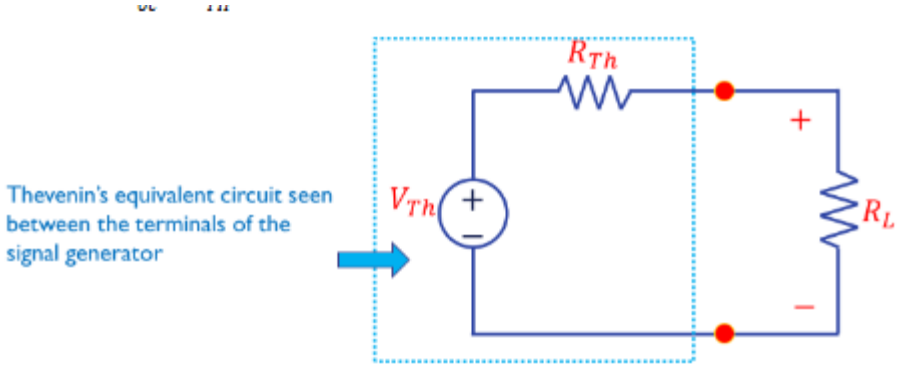


Figure 3: Thevenin equivalent circuit seen at the terminals of the signal generator, reproduced from the lab manual [1].

The measured open-circuit voltage (Voc) was 1.05V. The loaded voltage across the resistance load was 0.720V. Using the divider relationship $V_{RL} = Voc \times (R_L / (R_{TH} + R_L))$, the Thevenin resistance is $R_{TH} = R_L \times (Voc - V_{RL}) / V_{RL}$. R_L was taken as 99.12 Ohms, the experimental value of the 100 Ohm resistor. The calculated Thevenin resistance was 45.43 Ohms.

A 50 Ω load was then connected in place of the 100 Ω resistor, with the generator still set for a 1 Vpp signal. The experimental resistance value was measured to be 46.67 Ohms. The measured voltage, using the oscilloscope, with the resistance load was 0.530V. To compare, the theoretical voltage value for the Thevenin circuit was calculated using the Thevenin resistance. Using the divider relationship $V_{RL} = Voc \times (R_L / (R_{TH} + R_L))$ and the corresponding values, $V_{RL} = 0.532V$. This agrees with our experimental value.

The signal on the oscilloscope was observed returning to Voc (1.05V) from 0.530V while the resistor was being disconnected. This was expected as removing the load removes the voltage-divider effect entirely, so the open circuit voltage can be seen again.

3. Loading Theory

If the oscilloscope is connected to the function generator, the same voltage-divider relationship gives the fractional change in the open-circuit voltage: $\Delta V / Voc = RTh / (RTh + Rin)$.

Using $R_{TH} = 45.43$ Ω and Rin = 0.937 MΩ $\Delta V/Voc = 45.43 / (45.43 + 937{,}000) = 4.85 \times 10^{-5}$, or about 0.0049%. This confirms that connecting the oscilloscope to the function generator barely loads it at all, the change is negligible.

For a voltmeter or oscilloscope to not affect the voltage being measured by more than 1%, the same relationship requires $RTh / (RTh + Rin) \leq 0.01$, which rearranges to $Rin \geq 99 \times RTh$.

For our measured $R_{TH} = 45.43\Omega$, this threshold is $Rin \geq 99 \times 45.43 \approx 4500\Omega$, or about 4.50 kΩ. The oscilloscope's actual input resistance (0.937 MΩ) is roughly 200 times larger than this minimum, so the ≤1% loading condition is comfortably satisfied.

III. Discussion and Conclusions

This lab modeled two pieces of bench equipment as Thevenin equivalents and demonstrated measurement loading. The oscilloscope behaved like a resistor that ranged widely from hundreds of kilohms to a few megohms, while the function generator behaved like a low Thevenin resistance source, so connecting the two together caused very little loading, consistent with the 99× rule derived in Part III.

In our own measurements, the two oscilloscope-resistance methods did not agree well: the ohmmeter reading (0.2059 MΩ) was roughly 4.5 times smaller than the voltage-division result (0.937 MΩ), most likely because the oscilloscope was not fully isolated from the rest of the circuit when the ohmmeter reading was taken. The function-generator model fared much better: the 50 Ω load prediction (0.532 V) was within about 1% of the observed value (0.530 V), and the voltage correctly returned to Voc once the load was removed. Finally, with R_{TH} = 45.43 Ω and Rin = 0.937 MΩ, the ≤1% loading condition (Rin ≥ 99 × RTh) was satisfied by a wide margin, confirming that the oscilloscope does not meaningfully affect the function generator's output.

Overall, this lab showed how a complicated piece of equipment can be reduced to a simple voltage source and resistance, and how that simplified model is enough to predict loading behavior without needing to know the internal circuitry of the instrument.

IV. References

[1] Dr. Iman Salama. “Lab 5 – Thevenin Equivalents of Lab Equipment.” Northeastern University. 23 September 2023.